

Differential Active-S-box Analysis of the Krakken-2048 SPN Core: A Verified Two-Round Lower Bound and Its Limitations

Working note

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Abstract

This note records the results of an automated active-S-box analysis of Krakken-2048, carried out with a byte-lane Mixed-Integer Linear Programming (MILP) model, covering both *differential* and *linear* trails and both the SPN+XRBD subset and the complete round function. For the full (SPN+XRBD) round we prove the minimum differential active-S-box count at every round from one to eight: 5, 37, 69, 101, 133, 165, 197, 229. Each figure was obtained with the branch-and-bound optimality gap closed to zero (a proof, not a time-limited estimate), and the two-round SPN-core trail was independently verified layer-by-layer. The complete-round model (additionally including the PRESSURE/ARX and InkCloud layers) and the linear-trail model are proven to give the same counts at every round from one to eight—the expected behaviour for a wide-trail design with a maximal MDS branch number, where differential and linear active-S-box counts coincide and the post-Chi mixing layers cannot lower the floor set by XRBD.

The headline consequence: the full eight-round permutation has **no differential characteristic with fewer than 229 active S-boxes**, bounding any eight-round differential characteristic probability by 2^{-1374} and thereby ruling out single-characteristic differential cryptanalysis with very large margin; the linear bound, where confirmed, gives the corresponding correlation bound 2^{-1374} via the S-box maximum squared correlation of 2^{-6} . From two rounds onward each additional round adds exactly 32 active S-boxes. We remain explicit about scope: the per-S-box bounds use the standard active-S-box argument (not a cluster/hull analysis), and the PRESSURE layer is modelled conservatively (modular addition as activity-spreading, ignoring carries), which keeps all reported counts sound lower bounds. We also record a methodological finding: the open-source solver SCIP closes these instances where CBC’s lower bound stays frozen, making proven bounds attainable without a commercial solver.

1 Scope and methodology

We model differential propagation through Krakken-2048 at *byte-lane* granularity: the 2048-bit state is $32 \text{ words} \times 8 \text{ byte-lanes} = 256 \text{ lanes}$, one binary activity variable per lane (equivalently, one per S-box). A MILP minimises the total number of active S-boxes over N rounds subject to each layer’s activity-propagation constraints. When the solver reports an optimum *with a closed branch-and-bound gap*, the objective is a proven lower bound on the number of active S-boxes for any nonzero differential characteristic over those N rounds.

Why byte-lane and not word level. An earlier word-level model (one variable per 64-bit word) yielded a flat 4 active words per round, because at word granularity the bit-rotations (ρ , the XRBD rotations, the ARX shifts) are invisible: a rotation maps an active word to an active word and nothing more. Byte-lane granularity captures the essential effect that a rotation by

$r \not\equiv 0 \pmod{8}$ spreads an active lane across two adjacent output lanes, which is what makes activity compound across rounds.

Conservative (sound) modelling. Every modelling choice under-counts activity, so the resulting bound is a valid *lower* bound:

- Rotations are modelled as activity *spread* (an output lane is active if any input lane rotating into it is active).
- The MDS layer is modelled by its branch-number constraint: for each row and lane, the count of active input bytes plus active output bytes is either 0 or at least 9.
- Θ is modelled as non-cancelling column-parity spread.
- χ is modelled with its serial cross-coupling and the 32-bit half-swap.

MDS provenance. The 8×8 circulant MDS uses the Whirlpool coefficient row $(1, 1, 4, 1, 8, 5, 2, 9)$ over $\mathbb{F}_{2^8}$ with reduction polynomial `0x11D`, which has a proven branch number of 9 (the maximum for an 8×8 MDS matrix). We rely on this externally-established property rather than re-deriving it; we do, however, verify below that the matrix *as implemented* enforces branch 9 inside the round.

2 Result: a verified two-round bound

Rounds	Diff. SPN+XRBD (active S-boxes)	Diff. all layers (active S-boxes)	Linear all layers (active S-boxes)	DP / corr ² bound
1	5	5	5	2^{-30}
2	37	37	37	2^{-222}
3	69	69	69	2^{-414}
4	101	101	101	2^{-606}
5	133	133	133	2^{-798}
6	165	165	165	2^{-990}
7	197	197	197	2^{-1182}
8	229	229	229	2^{-1374}

Table 1: Minimum active-S-box counts, all entries proven to optimality (solver gap closed to 0). “Diff. SPN+XRBD” is the differential bound for $\Theta \rightarrow \text{MDS} \rightarrow \text{Rho} \rightarrow \text{Pi} \rightarrow \text{Chi} \rightarrow \text{XRBD}$. “Diff. all layers” and “Linear all layers” additionally include the PRESSURE (ARX) and InkCloud layers. All three columns coincide at every round—the expected behaviour for a wide-trail design with a maximal MDS branch number: differential and linear active-S-box counts agree, and the post-Chi mixing layers do not lower the count set by XRBD. The bound column is $(2^{-6})^{\text{count}}$: the differential-characteristic probability bound and, by wide-trail symmetry, the linear correlation-squared bound. For reference the SPN core alone (no XRBD) gives 5, 27, 55 at rounds one to three.

Headline result. The eight-round count of 229 active S-boxes proves that no eight-round differential characteristic has probability above 2^{-1374} . This is proven for the complete round function (all layers) and, by the linear model, the same bound holds for linear cryptanalysis (correlation-squared $\leq 2^{-1374}$). For an eight-round characteristic to threaten a 2048-bit-state construction its probability would need to be far higher; the proven bound is below this by a margin of more than a thousand bits, so single-characteristic differential and linear cryptanalysis are both ruled out conclusively. These are direct proofs at eight rounds, not extrapolations.

Differential/linear coincidence and the all-layers model. At every round from one to eight, the complete-round differential bound and the linear bound equal the SPN+XRBD differential bound (proven to optimality). This is the expected behaviour of a wide-trail design built on a maximal-branch (9) MDS matrix: at the byte-lane activity level the linear model is the transpose of the differential model, and the MDS branch number is identical for a matrix and its transpose, so the two active-S-box counts coincide; and the PRESSURE/InkCloud layers, being additional mixing, cannot lower the active-S-box floor that XRBD already forces. We therefore present the coincidence as a *confirmation* of wide-trail behaviour rather than as an independent rediscovery: the linear and complete-round models share the activity-level structure of the differential model, viewed from a different direction.

The linear trend and the XRBD contribution. From two rounds onward each additional round adds *exactly* 32 active S-boxes. At one round the SPN-core and full-round counts coincide (5), as expected—the S-boxes are counted at the Chi input, before the post-Chi layers act. Beyond one round the full-round count exceeds the SPN core by a growing margin (+10 at two rounds, +14 at three), confirming that XRBD’s cross-word rotations break the cheap lane-aligned trail the SPN core admits (a trail that rides the MDS at its exact branch number along a single byte-lane; see the Remark below). The flat +32 increment over eight proven data points is a strong empirical regularity, but we emphasise it is observed, not derived: the eight-round value rests on a direct proof, and we make no theoretical claim that the increment would remain 32 beyond eight rounds.

2.1 Independent verification of the trail

Because the bound is only as trustworthy as the encoding, the optimal trail was extracted and re-checked outside the MILP. Each layer was re-simulated in independent code on the solved input pattern and compared against the solver’s layer variables; all layers matched. The branch-number property was checked on every active MDS row-lane.

Minimal trail structure. The minimum-weight input difference has 4 active bytes, all in the same byte-lane (lane 7), located in words 0, 12, 19, 28. Its propagation is:

- **Round 0:** the 4 active bytes lie in 4 distinct MDS rows; each active row realises the branch number *exactly* (8 active inputs, 1 active output, sum 9), yielding 9 active S-boxes.
- **Round 1:** activity spreads to 16 active MDS row-lanes, again each at exactly branch 9 (8-in, 1-out), yielding 18 active S-boxes.

The total is $9 + 18 = 27$, matching the solver objective. Every active MDS row-lane satisfied $\text{in} + \text{out} \geq 9$ with zero violations, confirming the Whirlpool branch number is enforced by the implementation.

Remark 1. *The minimal trail rides the MDS at its exact branch number throughout and remains aligned to a single byte-lane. The per-round MDS activity grows as $9 \rightarrow 18$, i.e. linearly (factor 2) at the cheapest corner of the diffusion, rather than super-linearly. This is precisely the class of structured, position-aligned trail that the XRBD butterfly layer is intended to disrupt.*

3 What this does *not* establish

We state the limitations plainly, as each bears directly on whether the 8-round parameter is justified.

1. **Linear bound is structurally coincident, not independent.** The linear-trail MILP (the transpose of the differential model) is proven at all eight rounds and reproduces the differential count exactly, as wide-trail theory predicts. Because the activity-level linear model shares the differential model’s structure, it *confirms* rather than independently re-derives the bound; a genuinely independent check would require a bit-level mask model with explicit transposed linear maps (future work).
2. **Single characteristic, not differentials/hulls.** The argument bounds the best single trail. Differential attacks can in principle exploit clusters of characteristics, and linear attacks linear hulls. The 2^{-1374} margin leaves so much headroom that cluster/hull effects are extremely unlikely to bridge it. An empirical differential-clustering test on the minimal-trail input difference (companion experiments note, 200,000 pairs per round, 128-bit output projection) found no detectable cluster at any round from one to eight (detection floor $\approx 2^{-17}$), giving direct evidence against a fat cluster on this trail; a full analytical cluster/hull quantification across all input differences remains open.
3. **PRESSURE modelled conservatively.** The complete-round model includes the PRES-SURE (ARX) layer with modular addition modelled as activity-spreading (carry ignored). This can only *add* activity, so the reported counts remain sound lower bounds; a carry-aware ARX model could only raise them.
4. **The flat +32 increment is observed, not proven in general.** Every value through eight rounds is a direct proof, so the eight-round figure does not depend on the trend. We make no claim that the per-round increment remains 32 beyond eight rounds.

Net assessment. On the differential metric the eight-round parameter is directly justified: 229 active S-boxes and a 2^{-1374} characteristic bound place the design far beyond the reach of differential cryptanalysis. The complete-round model (all layers) and the linear model are proven to give the same counts at every round from one to eight, extending the conclusion to the full round function and to linear cryptanalysis under the wide-trail coincidence described above. The remaining refinements—a bit-level independent linear model, a carry-aware ARX model, and cluster/hull estimates—are expected to preserve or strengthen the picture, not weaken it.

4 Methodological note: solver choice is decisive

The bounds above were not obtainable with the default open-source MILP solver. With CBC, the lower bound (“best possible”) stayed frozen at the LP-relaxation value $73/512 \approx 0.143$ for the entire run at three rounds: the solver found feasible trails (upper bounds $57 \rightarrow 56 \rightarrow 55$) but generated no effective cutting planes, so the gap never closed and the run terminated on its time limit with a meaningless lower bound. The two-round instance was provable under CBC only because its branch-and-bound tree was small enough to exhaust by enumeration.

SCIP, by contrast, generated tens of thousands of cuts and moved the dual bound steadily ($0.14 \rightarrow 8 \rightarrow 22 \rightarrow 35 \rightarrow 55$), closing the three-round gap to 0 in roughly 51 minutes on a single core. We also tested a hand-added convex-hull MDS lifting ($\sum_{\text{row}} v \geq 9v_k$ for each variable); it preserved correctness (two-round result unchanged at 27) but did *not* accelerate convergence—it left the root LP relaxation essentially unchanged because that relaxation’s weakness is a uniform-fractional-activity artifact spanning all lanes, not a per-row deficiency. The practical conclusion is that a strong cut-generating solver (SCIP, or commercial alternatives) is the decisive ingredient for these instances; formulation tightening of the per-row MDS constraint alone is not.

5 Reduced-round distinguisher experiments

The permutation was subjected to two classic structural distinguisher tests using a reduced-round harness (`harness.c`) driven by the `run.sh` build-and-run wrapper. Both tests operate directly on the *permutation* (not the sponge) at a configurable round count and use a deterministic splitmix64 PRNG seeded with the round count for reproducibility.

Test 1: Integral / saturation distinguisher. A structure of $2^8 = 256$ inputs is formed by saturating one chosen input byte over all 256 values while holding every other byte constant at a random value. The reduced-round permutation is applied to each element and all 256 outputs are XOR-summed. For an ideal permutation, each output byte is balanced (XOR-sum = 0) with probability $1/256$ by chance, giving an expected baseline of *structures*/256 zero-sum events per output byte position across many random structures. A byte position that is zero far more often than this baseline—specifically, more than $8\times$ the ideal *and* at least 4 hits—is flagged as a possible distinguisher.

Test 2: Zero-sum / cube distinguisher. A d -dimensional input subspace is formed by choosing d random bit positions in the 2048-bit state and sweeping all 2^d combinations of those bits (holding the remaining bits constant at a random value). The outputs over the full cube are XOR-summed. A sum of zero across *all* 256 output bytes constitutes a zero-sum distinguisher. Multiple random cube placements are tried per round and the best (most zero output bytes) is reported.

Results. Table 2 records all runs. At every round count from 1 to 4, both tests are consistent with the random-permutation baseline. No integral distinguisher was detected at any round: the highest observed ratio of balanced bytes to the ideal baseline was $1.8\times$ (rounds 3 and 4), well below the $8\times$ detection threshold. No zero-sum distinguisher was detected: the best zero-sum result across all trials was 4/256 output bytes summing to zero (round 4, cube dimension 18), against an ideal expectation of approximately 1.

Rounds	Structures	Integral best ratio	Cube dim d	Cube trials	Zero-sum best
1	4096	$1.6\times$ ideal	10	8	3/256
2	4096	$1.7\times$ ideal	10	8	2/256
3	4096	$1.8\times$ ideal	14	16	3/256
4	4096	$1.8\times$ ideal	18	16	4/256

Table 2: Reduced-round integral and zero-sum distinguisher results. “Integral best ratio” is the highest observed ratio of balanced-byte events to the ideal baseline ($\text{structures}/256 = 16$) across all 256 output byte positions; the flagging threshold is $8\times$. “Zero-sum best” is the maximum number of output bytes that summed to zero across all cube placements in that run; the ideal expectation for a random permutation is ≈ 1 . No distinguisher was detected at any round count. The cube dimension was increased with the round count to compensate for the growing algebraic degree: 2^{18} inputs per trial at round 4.

Interpretation. The integral result confirms that no output byte is forced to balance by the permutation structure at any of the tested round counts: even at one round, the XOR-sum distribution over saturated structures is consistent with a random bijection. The zero-sum result is consistent with the algebraic degree growing rapidly with the round count: a zero-sum distinguisher requires a cube of dimension at least equal to the algebraic degree of the output

in the chosen input variables, and no such cube was found up to dimension 18 at four rounds ($2^{18} = 262,144$ evaluations per trial, 16 independent placements). These results complement the MILP active-S-box bounds: the MILP bounds rule out differential and linear distinguishers analytically; the harness results give direct empirical evidence against integral and algebraic zero-sum distinguishers at reduced round counts.

Division-property integral (analytical, MILP). The two tests above are empirical samplings. Their analytical counterpart is the bit-based division property (two-subset CDBP), which proves an output bit balanced exactly when its division trail through the round function is infeasible—a proof, not a statistical observation. We have built a sound MILP harness for this: each round is modelled bit-for-bit (the linear layers by their exact $\text{GF}(2)$ matrices, the ARX PRESSURE layer by a sound full-adder over-approximation, the ABYSSAL S-box by its division transitions), with the circuits validated bit-for-bit against the compiled permutation. The practical obstacle was the S-box model: the standard “selector” encoding adds ≈ 2015 binary variables per S-box ($\approx 5 \times 10^5$ per round), and at that size the solver could not resolve a single output bit. Replacing it with a compact, *exact* inequality model (661 inequalities, no auxiliary variables, verified over all 2^{16} points of the S-box division polytope) cuts the one-round program from 937,952 to 422,112 variables and makes it solvable: round 1 now solves to optimality (output bit 0 proved reachable in ≈ 160 s) where the selector model never terminated. The round-1 result is consistent with the empirical integral finding—first-round diffusion already balances away—and the exhaustive multi-round sweep that would establish the integral-distinguisher length is an ongoing computation. The harness is released with this analysis for reproduction and extension.

6 Rotational and RX-difference distinguisher experiments

The one-round permutation was probed for rotational and RX-difference structure using a dedicated command-line tool (`rotational`), run in heavy mode (`-heavy -no-timeout`) with 500 000 trials per configuration and the timeout disabled. Four test categories were executed.

Test 1: Word-level rotational distinguisher. Pairs of inputs related by a rotation of $r \in \{1, 2, 3, 4\}$ words were fed to the one-round permutation and the fraction of output pairs preserving the same rotation was recorded. For all four rotation offsets, the number of consistent output words was 0 against a random baseline of $\approx 3.47 \times 10^{-13}$. No word-level rotational symmetry was detected.

Test 2: Bit-level rotational distinguisher. The same procedure was applied at bit granularity for rotation offsets $r \in \{1, 7, 13, 19, 32\}$ bits. All five offsets yielded 0 consistent output words. No bit-level rotational symmetry was detected.

Test 3: Single-word RX-difference distinguisher. For rotation constants $rx_rot \in \{7, 13, 17, 19, 31, 37\}$ and injection words $\{0, 8, 16\}$, the maximum per-output-bit bias across all 2048 output bits was measured over 500,000 trials. The 5σ detection threshold at this sample size is 0.003536. In 17 of the 18 configurations the maximum bias was well below threshold (range 0.002228–0.002796).

One configuration (`rx_rot=17`, `inj_word=8`) yielded a maximum bias of 0.003572 at bit 1421, flagging 1 out of 2048 bits above the 5σ threshold. Given the number of bits tested across the full suite, a single marginal exceedance at $\approx 1.01 \times$ the threshold is consistent with the expected false-positive rate under the null hypothesis and is not considered a genuine distinguisher candidate; see the heavy-mode sweep below for independent confirmation.

Test 4: Heavy-mode PRESSURE-block RX sweep. A systematic sweep over all 63 non-trivial rotation constants was conducted for four multi-word injection configurations: `blkA` (words 0–3), `blkB` (words 16–19), `blkA+B` (words 0–3 and 16–19), and `w0-7` (words 0–7), each at 500,000 trials per rotation constant (252 jobs in total; wall-clock time ≈ 962 s). The five highest-ranking results are shown in Table 3. No configuration exceeded the 5σ threshold, and the highest bias observed across the entire sweep was 0.003472—1.8% below the 5σ threshold of 0.003536.

Rank	Config	rot	max bias	5σ thresh	Anomalous bits	Top bit
1	<code>blkB(w16-19)</code>	48	0.003472	0.003536	0	1873
2	<code>w0-7</code>	21	0.003198	0.003536	0	1024
3	<code>blkA+B(w0-3,16-19)</code>	49	0.003162	0.003536	0	1883
4	<code>w0-7</code>	14	0.003146	0.003536	0	973
5	<code>blkB(w16-19)</code>	12	0.003140	0.003536	0	442

Table 3: Top-5 results from the heavy-mode RX sweep over all 63 rotation constants and 4 multi-word injection configurations (500,000 trials each). No configuration produced any output bit with bias above the 5σ threshold of 0.003536. The highest observed bias of 0.003472 is 1.8% below the detection threshold.

Interpretation. All four test categories return results consistent with the random-permutation null hypothesis. The marginal single-bit exceedance observed in the single-word sweep (Test 3) is not reproduced in any of the 252 independent configurations of the heavy sweep (Test 4), confirming it as a statistical fluctuation rather than a structural anomaly. At one round, no rotational or RX-difference distinguisher is detectable at the tested trial counts and significance level.

7 Empirical cryptanalytic probe suite

The permutation was subjected to a five-test empirical probe suite (`attacks.c`), targeting attack families not directly addressed by the MILP active-S-box analysis: impossible differentials, algebraic degree growth, boomerang distinguishers via the BCT, differential-linear bias, and invariant subspace / S-box coset structure. Tests were run at one and two rounds in normal mode (timeouts enabled) with the default trial budgets. Results are summarised in Table 4; the subsections below discuss each test in detail.

7.1 Test 1: Impossible differential

For each of four input-bit positions (0, 64, 512, 1984) and four output-byte positions (0, 31, 127, 255), 200,000 random pairs $(x, x \oplus \delta)$ were encrypted and the output-difference histogram at the chosen output byte was recorded. Any output difference with a histogram count of zero constitutes an impossible-differential candidate.

At one round all completed input-bit configurations were clean (0/255 non-trivial differences missing); the 30-second timeout interrupted the run before the final input bit (`input_bit` = 1984) could be completed. At two rounds the timeout fired after two of the four input bits, again with no candidate found. Neither run produced an impossible-differential candidate.

7.2 Test 2: Algebraic degree growth

The multilinear degree of chosen output bits over $\text{GF}(2)$ was measured using the top-coefficient (higher-order derivative) method. The test climbs dimension d from 1 upward, running 5,000

Test	Attack family	1 round	2 rounds
1	Impossible differential	Clean	Clean
2	Algebraic degree growth	≥ 23	≥ 21
3	BCT / boomerang	BCT max = $6^\dagger$; no full-cipher distin- guisher	(same)
4	Differential-linear	Skipped ($r < 2$)	Clean; max $ \text{corr} = 0.0034$
5	Invariant subspace	Clean	Clean

Table 4: Summary of empirical probe results at one and two rounds. [†] The BCT maximum of 6 is a property of the S-box alone and is round-independent; it does not translate to a full-permutation distinguisher at any tested round count.

inner trials per step; each step at dimension d costs 2^d permutation evaluations per trial. It stops at the first d where no nonzero top coefficient is found, reporting the last confirmed d as an empirical lower bound.

At one round, all inner trials returned a nonzero top coefficient at every d from 1 to 23; the run was cut by the 120-second timeout before completing $d = 24$. At two rounds the confirmed lower bound is ≥ 21 , with the timeout firing at $d = 22$ (two-round evaluations are more expensive).

Both results are consistent with the theoretical expectation: the full-state algebraic degree saturates to approximately 2048 within one to two rounds, owing to the S-box’s per-byte degree of 7 combined with the diffusion layers. The timeouts are not failures; they reflect the exponential cost of reaching high d with the default 120-second cap rather than any absence of degree.

7.3 Test 3: S-box BCT and boomerang distinguisher

Part A: Boomerang Connectivity Table. The full 256×256 BCT of the Abyssal S-box was computed. The maximum entry (excluding $(a, b) = (0, *)$ and $(*, 0)$) is 6, occurring at $(a, b) = (0x01, 0x05)$. The distribution of non-zero entries is: 6 at 510 pairs, 4 at 255 pairs, 2 at 31,620 pairs.

A BCT maximum of 6 exceeds the DDT-flat ideal of 2 (probability 2^{-7} per switch), giving a per-S-box boomerang switch probability of $2^{-5.42}$. This is a property of the S-box alone and is round-independent. It warrants noting but, as Part B shows, does not propagate to a full-permutation distinguisher.

Part B: Empirical boomerang probe. A 50,000-trial empirical boomerang quartet experiment was run at one and two rounds using input difference $\alpha = 0x01$ at byte 0. At one round the maximum output-difference count was 232 against a uniform expectation of 195.3 (ratio 1.19); at two rounds the count was 229 (ratio 1.17). Both are well below the 2.0 anomaly threshold. No full-permutation boomerang distinguisher was detected at either round count.

7.4 Test 4: Differential-linear distinguisher

This test requires at least two rounds to split the permutation into a top and bottom half; it was skipped at one round. At two rounds, five configurations of (r_0, δ, γ) —input difference byte, output bit mask, and split—were probed over 200,000 trials each. The 5σ bias threshold at this sample size is 0.011180. The highest observed correlation was $|\text{corr}| = 0.003380$ ($r_0 = 1$, $\delta = (\text{byte } 0, 0x01)$, $\gamma_{\text{bit}} = 0$), less than a third of the detection threshold. All five configurations were below the threshold; no differential-linear bias was detected.

7.5 Test 5: Invariant subspace and S-box coset check

Part A (dimension-1 S-box cosets). An exhaustive enumeration of all (a, g) pairs satisfying $S(a) \oplus S(a \oplus g) = g$ found 282 hits across 141 distinct generators (out of 255). These are the candidate seeds of a dimension-1 invariant subspace trail.

Parts B and C (full-round survival). None of the 282 hits survived a full round in the constant-base probe (Part B), nor in the random-base empirical probe (Part C, 2,000 trials per sampled pair). Both tests were run at one and two rounds with identical conclusions: the S-box coset structure does not extend through a full round of the permutation.

Part D (dimension-2 exhaustive check). All 32,385 dimension-2 subspaces were checked exhaustively; no coset hit was found. The invariant trail therefore cannot be extended to higher dimension via the S-box layer.

The 282 dimension-1 hits are not unusual for an 8-bit S-box and do not constitute a weakness: the decisive test is whether the structure survives the full round function, and it does not.

8 Bit-level diffusion: SAC, completeness, and monobit

The structural sanity checks below report *mean* avalanche (a single input flip changes ≈ 0.5 of the 2048 output bits). A correct mean is necessary but not sufficient for good diffusion: it is compatible with a structured per-bit dependency in which an input bit deterministically flips some output bit and never flips another. We therefore measured the full strict-avalanche-criterion (SAC) matrix (Webster and Tavares, 1986),

$$\text{SAC}[i][j] = \Pr[\text{output bit } j \text{ flips} \mid \text{input bit } i \text{ flipped}], \quad \text{ideal } \frac{1}{2},$$

over all $2048 \times 2048 = 4.19 \times 10^6$ (input,output) bit pairs, estimated with $N = 4096$ random base states per input bit at each round count from one to eight (run against the AVX2 permutation; $2 \times 2048 \times N$ permutation evaluations per round). From the same counters we extract *completeness* (each cell strictly between 0 and N : every input bit influences every output bit, none deterministically) and a per-output-bit *monobit* balance over random inputs. At $N = 4096$ the two-sided 5σ bound on a single proportion is $|p - \frac{1}{2}| > 0.0391$.

At every round, including round 1, the SAC matrix is statistically indistinguishable from ideal: the mean cell deviation equals the random-sampling value and the few cells beyond 5σ match the expected false-positive count over 4.19×10^6 cells. The completeness result is the decisive one—with no dead and no stuck cells at any round, every one of the 2048 input bits provably influences every one of the 2048 output bits, and none does so deterministically. This is the per-bit form of the full-state first-round diffusion that the mean-avalanche figure only asserts on average, and a design with a correct mean but a structured dependency matrix would have failed it. No output-bit monobit bias was detected at any round. The test reuses the deterministic splitmix64 seeding convention of the other harnesses (seeded by the round count) and is reproducible.

9 Statistical randomness of the sponge keystream (NIST SP 800-22)

The active-S-box bounds and the bit-level diffusion results above concern the permutation directly. As a complementary mode-level check we ran the NIST Statistical Test Suite (SP 800-22) on the keystream produced when the permutation is used in the stream-cipher mode of the Krakken-Disk volume encryptor, where $\text{ciphertext}_i = \text{plaintext}_i \oplus \text{Sponge}(\text{key} \parallel \text{nonce} \parallel \text{LE64}(i))$.

Round	SAC max $ p - \frac{1}{2} $	SAC mean $ p - \frac{1}{2} $	cells $> 5\sigma$ (exp. ≈ 2.4)	dead / stuck cells	monobit max $ f - \frac{1}{2} $
1	0.0415	0.0062	4	0 / 0	0.0276
2	0.0405	0.0062	2	0 / 0	0.0295
3	0.0422	0.0062	5	0 / 0	0.0305
4	0.0422	0.0062	2	0 / 0	0.0249
5	0.0413	0.0062	3	0 / 0	0.0264
6	0.0427	0.0062	1	0 / 0	0.0293
7	0.0435	0.0062	4	0 / 0	0.0271
8	0.0413	0.0062	4	0 / 0	0.0303

Table 5: SAC matrix, completeness, and monobit balance ($N = 4096$ per input bit; $5\sigma = 0.0391$). The mean cell deviation 0.0062 is exactly the value expected of an ideal Binomial($N, \frac{1}{2}$) proportion ($\approx 0.399\sigma$); the 1–5 cells per round exceeding 5σ track the ≈ 2.4 false positives expected over 4.19×10^6 cells. Zero dead and zero stuck cells at every round establish full bit-level completeness.

Three independent 5500MB volume containers were generated empty, so the plaintext is the all-zero sector image and the resulting ciphertext is—up to per-sector random nonces (24 B per 4096 B sector) and a small fixed header—the raw Krakken sponge keystream. The randomness test therefore reflects the permutation output, not any plaintext structure.

Each file was assessed over 10 sequences of 10^6 bits. All fifteen STS test families passed for all three files, each with a proportion of passing sequences at or above the NIST minimum (8/10, or 5/6 for the random-excursion tests) and non-degenerate uniformity P -values: Frequency, Block Frequency, Cumulative Sums, Runs, Longest Run, Binary Matrix Rank, Spectral (FFT), Non-Overlapping Template (148 templates), Overlapping Template, Maurer’s Universal, Approximate Entropy, Random Excursions and Variant, Serial, and Linear Complexity. A custom high-throughput monobit and runs test over a larger 10^8 -bit window of each file returned unremarkable P -values (monobit 0.72, 0.22, 0.99; runs 0.53, 0.31, 0.53 for the three files) and a byte-level Shannon entropy above 7.999 bits/byte in every case.

This is an empirical, mode-level result: like the distinguisher battery it is necessary but not sufficient, excluding detectable first- and second-order statistical bias in the keystream at the sampled scale rather than establishing a property of the volume-encryption construction itself.

10 Sponge-hash collision and output-distribution battery

Complementing the keystream randomness tests, the sponge *hash* (256-bit digest, rate 160 bytes) was probed for collision and distribution behaviour—a family neither the active-S-box analysis nor the distinguisher battery addresses—with three quick tests against the compiled hash.

Birthday collisions. Hashing $N = 10^6$ random messages, truncating each digest to t bits, and counting colliding pairs reproduces the ideal random-function expectation $\binom{N}{2}/2^t \approx N^2/2^{t+1}$ (Table 6). Observed counts match expectation closely wherever it is statistically meaningful ($t \leq 34$: 1835 vs 1863, 495 vs 466, 122 vs 116, 27 vs 29); at $t \geq 36$ the expectation falls below 7 and the ratios are small-count Poisson noise. No truncation width shows a collision excess.

Output diffusion / near-collisions. For 2×10^5 message pairs differing in one input bit, the 256-bit digest difference had mean Hamming weight 127.98 (ideal 128), standard deviation 8.00 (ideal 8), minimum 93 and maximum 165. The minimum far from 0 excludes cheap one-bit near-collisions.

t (bits)	expected	observed	obs/exp
28	1862.6	1835	0.985
30	465.7	495	1.063
32	116.4	122	1.048
34	29.1	27	0.928
36	7.28	5	0.687
38	1.82	2	1.100
40	0.46	1	2.199

Table 6: Truncated-digest birthday collisions, $N = 10^6$ messages, against the ideal $N^2/2^{t+1}$.

Digest byte uniformity. Over 6.4×10^6 output bytes the byte-value χ^2 was 273.8 on 255 degrees of freedom (ideal 255 ± 22.6 ; $z = +0.83$), consistent with a uniform output.

All three are empirical and sampling-bounded: they exclude gross collision or distribution defects at the $\sim 10^6$ -message scale, not establish collision resistance analytically.

11 Remaining work

The differential question that motivated this analysis is settled: the complete round function has a proven minimum of 229 active S-boxes at eight rounds, for both differential and linear cryptanalysis. What remains, in priority order, strengthens the analysis but is not expected to change the conclusion:

1. **Independent bit-level linear model.** Replace the activity-level (transpose-coincident) linear model with an explicit bit-level mask model, for a check not sharing the differential model’s structure.
2. **Carry-aware ARX model.** Model PRESSURE’s modular addition with the standard addition inequalities rather than activity-spreading.
3. **Cluster / linear-hull estimates**, to complement the single-trail bounds.
4. **Independent verification and external cryptanalysis.** Publish the specification, the MILP models, and the proven optima (each solver run is reproducible) for third-party scrutiny.

Structural sanity checks

Independently of the active-S-box bounds, the permutation was subjected to empirical structural checks (not proofs): bijectivity on a sample (2000/2000 distinct outputs), absence of fixed points (0 among 503 states including the all-zero and all-one states), strict avalanche (single input-bit flip changes a mean of 0.5004 of the 2048 output bits, minimum 0.47), and a sampled subspace-trail collapse test (random low-dimensional affine subspaces show no rank collapse under the permutation). All checks pass. These rule out several gross structural weaknesses but are not a substitute for an analytical invariant-subspace / subspace-trail study, which remains future work.

Reproducibility

The analysis tooling accompanying this note comprises: a scalar/AVX2 equivalence harness (the two implementations are bit-identical on the permutation at all round counts and on the hash

across all tested input and output lengths); the byte-lane MILP bound prover (differential and linear; SPN-core, SPN+XRBD, and complete-round modes); the decomposed SPN-core prover; a single-lane propagation probe; the trail-extraction verifier used to confirm the two-round result and the branch-number property; and the structural sanity checker. All bounds were obtained with the SCIP solver; each run is reproducible and terminates with a certificate of optimality (or, where noted, a proven lower bound).